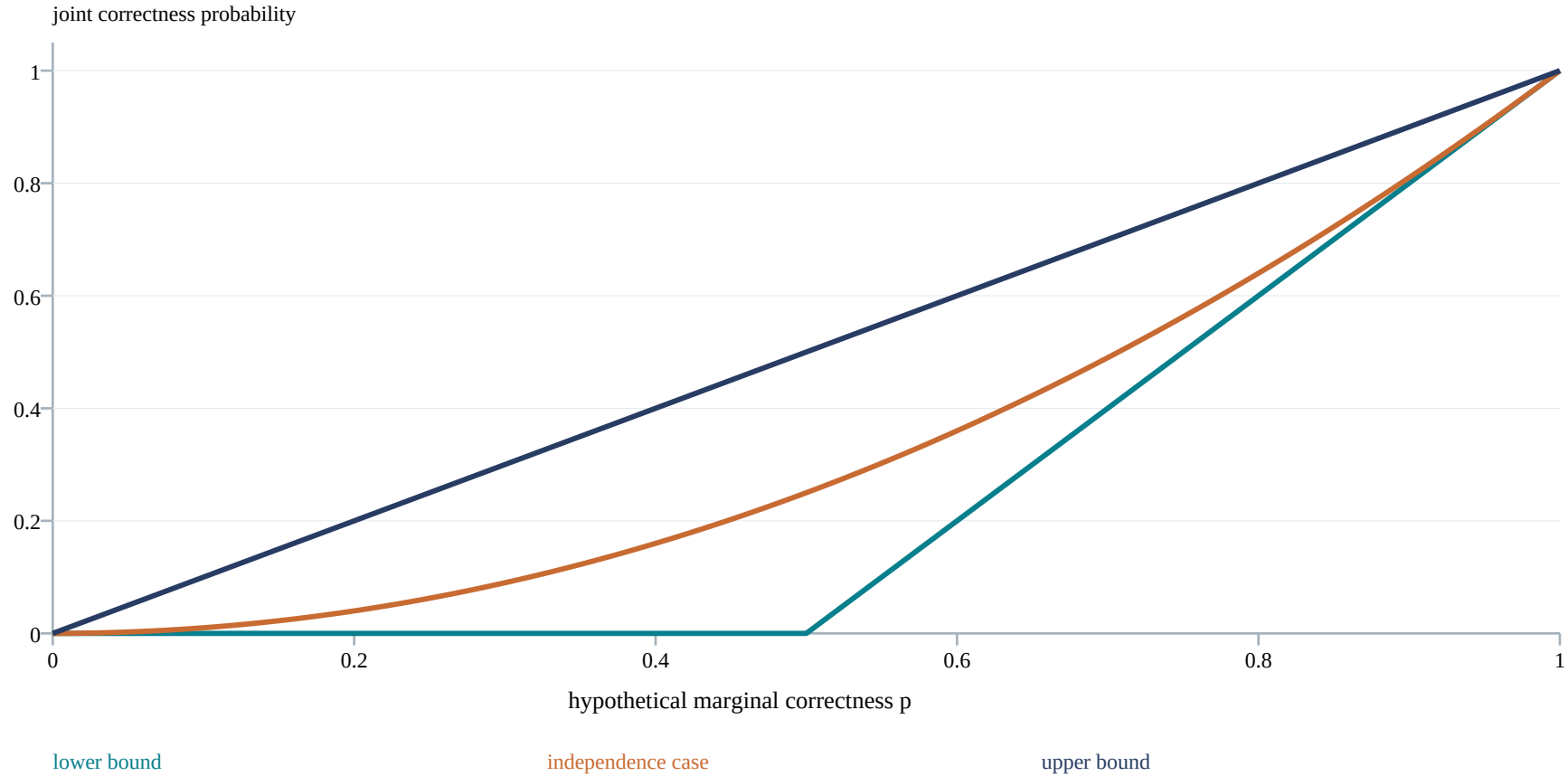
Marginals do not determine joint correctness

Analytic bounds; hypothetical equal marginal probability p, not API confidence